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The Injury Rate in NFL Players was Increased Following Cancellation of Preseason Games due to COVID 19

Running Title: NFL Injury Rate Increased due to COVID 19

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Abstract

Purpose: To investigate the injury rate in NFL athletes during the first four weeks of the 2020 NFL season.

Methods: This study was a retrospective review of all NFL players who were placed on the injury report during the preseason and first four weeks of the regular season from the 2016-2017 through 2020-2021 NFL regular seasons. Player's date of injury was cross-referenced with an absence of statistics from the respective games for which they were ruled out to ensure accuracy. Injury rates were calculated per 1000 athletic exposures (AE). Relative risk (RR) with 95% confidence intervals (CI) compared injury rates between the two cohorts.

Results: Over the course of the study period of four NFL seasons 3025 injuries were reported. Of the 3025 injuries reported, 582 (19%) occurred during weeks 1-4 of the 2020-2021 regular season, while 1292 (53%) occurred during preseason weeks 1-4 and 1151 (38%) during regular seasons weeks 1-4 of NFL seasons 2016-2017, 2018-2019, and 2019-2020. There was a significant increase in the injury rate during weeks 1-4 of the 2020-2021 regular season for all comparisons with the injury rate both during the preseason and regular season of three recent past NFL seasons.

Conclusions: The rate of injury in NFL players during weeks 1-4 of the 2020-2021 regular seasons was significantly higher than three recent past NFL preseasons and regular seasons.

Level of Evidence: Level IV, diagnostic case series.

Introduction

Due to rising number of positive Covid-19 tests observed in NFL players during the summer of 2020, the NFL suspended the 2020 preseason on July 21, 2020[1]. Suspension of play out of concern regarding rising cases of Covid-19 in the United States was widespread and not limited to the NFL, similar trends were observed across all sports and levels including high school collegiate, and professional. In order to mitigate potential Covid-19 exposures NFL team training

camp roster sizes were limited to 80 players[1]. Typically, the NFL preseason serves as the primary time for preparing athletes for the rigors of the upcoming NFL season. However, given this unprecedented stoppage of preseason training and teams inability to continue with practice and games due to social distancing precautions, athletes were forced to train and condition on their own. We speculated that athletes, in the absence of organized team activities, developed sports deconditioning.

Previous studies have demonstrated that fatigue and deconditioning may place athletes at a higher risk for injury. The International Olympic Committee identified absolute load, calendar congestion, and psychological load to be closely associated with injury risk through a variety of mechanisms.[2] Excessive training that exceeds the body's loadbearing capacity causes microtrauma to tendons, muscles, and bones, which is the etiology of overuse injuries.[3-8]

In light of these findings highlighting the association between fatigue and risk of injury, investigating modifiable causes influencing athletes' levels of fatigue is a critical step in reducing risk of injury.

The purpose of this study was to investigate the injury rate in NFL athletes during the first four weeks of the 2020 NFL seasons. Our hypothesis was that the observed injury rate during the first month of the 2020-2021 NFL season would be significantly higher when compared to past seasons.

Methods

Data collection

This study was a retrospective review of all NFL players who were placed on the injury report during the preseason and first four weeks of the regular season from the 2016-2017 through 2020-2021 NFL regular seasons. The data was compiled using publicly available injury reports (Appendix). The sources used publish a weekly injury reports detailing the name of the players injured, their corresponding teams, and a brief description of each injury. At least two different

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sources were used to cross-reference injury information. To verify each player's date of injury, we confirmed each player's reported injury date corresponded to an absence of statistics from the respective games for which they were ruled out. These methods are in accordance with previously published protocols.[9-13] When collecting data, we noted that the 2017-2018 records were incomplete, thus we excluded them from the study. We felt that data from three recent past seasons to compare data from 2020 to was adequate.

Team records available on individual team websites were used to collect demographic information. Our dataset included each player's name, position, team, pathology, age at injury, date of injury, season. Data for the 2020 season was collected prospectively, thus we were unable to determine games missed due to injury as the 2020 NFL season was ongoing when this manuscript was written.

Definitions

Reportable injuries were based on the criteria that the injury necessitated a team physician referral or emergent care and resulted in missed athletic activity. An athlete-exposure (AE) was defined as one athlete's participation in one game in which they were exposed to potential injury.

Data and Statistical Analysis

Data analysis was conducted using SPSS 25 (SPSS Inc, Chicago, Illinois). Injury rates were calculated per 1000 AEs using the following calculations; for the first 3 preseason games where NFL teams carry 90 players on the roster (Total number of injuries x 1000/ Total number of AEs (AEs = 90 players * 32 NFL teams * number of games)); for the fourth preseason game where NFL teams cut the roster to 75 players (Total number of injuries x 1000/ Total number of AEs (AEs = 75 players * 32 NFL teams * number of games)); and for the regular season games (Total number of injuries x 1000/ Total number of AEs (AEs = 53 players * 32 NFL teams * number of games)). Risk ratio (RR) with 95% confidence intervals (CI) compared injury rates between the two cohorts. Risk ratios were calculated comparing both weeks 1-4 from the preseason and regular season of NFL seasons 2016-2017, 2018-2019, and 2019-2020 to the 2020-2021 season. Week 1 of the 2020-2021 NFL season was compared to both past preseason and regular season week 1, this method of comparison was continued for weeks 2-4. Continuous variables were

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statistically compared using two-tailed t-test, while categorical variables were compared using Chi-squared test. Statistical significance was set at $P \leq 0.05$.

Results

Over the course of the study period of four NFL seasons 3025 injuries were reported. Of the 3025 injuries reported, 582 (19%) occurred during weeks 1-4 of the 2020-2021 regular season, while 1292 (53%) occurred during preseason weeks 1-4 and 1151 (38%) during regular seasons weeks 1-4 of NFL seasons 2016-2017, 2018-2019, and 2019-2020 (Table 1 and Figure 1). There was a significant increase in the injury rate during weeks 1-4 of the 2020-2021 regular season for all comparisons with the injury rate both during the preseason and regular season of three recent past NFL seasons (Table 2).

Positional breakdown demonstrated that wide receivers, safeties, and linebackers sustained the most injuries during the study period (Table 3). Most reported injuries were to the knee 473 (16%), hip/groin 347 (11%), and ankle 316 (10%) (Table 4). The majority of injuries occurred secondary to player contact 2151 (71%). There were 343 (11%) undisclosed injuries identified.

Discussion

The most important finding of this study is that the injury rate during weeks 1-4 of the 2020-2021 NFL season was significantly higher than the injury rates during weeks 1-4 of the 2016-2017, 2018-2019, and 2019-2020 NFL preseason and regular season. The findings of our study confirmed our hypothesis that players were at a higher risk of injury during the early 2020-2021 regular season following cancellation of preseason games due to Covid 19. Our findings highlight the importance of NFL training camp in preparing NFL athletes for the rigors of the NFL regular season and its influence on injury prevention.

Similar to the findings of Feeley et al.[14] and Elliott et al.[15] we observed more injuries occurred during the first two weeks of the NFL preseason 777 (60%) when compared with the second two

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weeks 515 (40%). This trend was noted to continue, less dramatically, during the 2020-2021 regular season in that 300 (52%) injuries occurred during the first two weeks while 282 (48%) occurred during the second two weeks. The results of our study suggest that in the absence of an NFL preseason injuries are more frequent during the early regular season, similar to the injury trend noted by Feeley et al.[14] during training camp. However, given the unprecedented nature of the Covid-19 pandemic and its effect on the 2020-2021 NFL season, this finding should be interpreted with this limitation in mind.

We speculate that similar factors as the ones proposed to account for the increased injury rate noted in the early NFL preseason also contributed to the increased injury rate noted in NFL players during weeks 1-4 of the 2020-2021 regular season. These factors include the relative deconditioning and muscle weakness that occurs during the offseason,[16,17] as well as fatigue, which suggests that fatigued muscles are more vulnerable to injury.[18,19] It is likely that the deconditioning of players during the early 2020 NFL regular season was similar to past NFL preseasons; however, during the 2020 season players were asked to play more time and at a higher level than they would typically be expected to play during the early preseason, specifically starters. Thus, increased time and intensity of play in the context of baseline deconditioning likely explains why the injury rate during the early 2020 NFL season was higher than the injury rate in past NFL preseasons. Our findings contribute to growing body of evidence that fatigue and deconditioning may place athletes at a higher risk for injury.

Saw et al. in their systematic review of subjective measures used to monitor an athletes response to training noted subjective measures including mood disturbance, perceived stress and recovery, and symptoms of stress, to be highly sensitive to an athlete's well-being in response to acute and chronic exercise loads.[20] In light of these findings the NFL may consider implementing athlete self-reported questionnaires in order to monitor athlete's level of exhaustion. Benefits of monitoring fatigue and athlete well-being include revealing the need for recovery in order to prevent injury, guiding training and competition scheduling, and understanding changes in athletic performance related to fatigue.[2,21] To our knowledge, no such measures have been practiced in attempt to reduce injury risk in professional football. Future investigations may be useful in evaluation of these subjective measurement techniques and their efficacy in injury reduction in the NFL.

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Limitations

This study, as well as other similar publications utilizing the well-established multistep protocol to identify injuries in professional athletes through review of public records, is associated with several limitations. This is a database study, and incomplete data entry and inaccurate reporting of injuries could confound the season-to-season injury differences. Details regarding injury diagnosis and management, including injury severity, exact pathology, imaging reports, and exact medical clearance were not available for all players. Because injured players were identified using public records, the possibility of reporting errors, and omissions exist. We did not assess the rate of injuries over the entire season to see if the injury rates normalized as the season went on. Our study also lacks objective measures of deconditioning to correlate with injury rates during COVID-19. Finally, there remains the possibility of selection bias in the injured group in which only "newsworthy" players injured were reported on. We attempted to minimize this bias by corroborating reported injuries with two additional resources, but the data is limited by what is publicly available.

Conclusions

The rate of injury in NFL players during weeks 1-4 of the 2020-2021 regular seasons was significantly higher than three recent past NFL preseasons and regular seasons.

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References

1. Pickman B. Report: NFL Scraps 2020 Preseason Amid COVID-19 Pandemic. *Sports Illustrated* July 21, 2020. Available at: https://www.si.com/nfl/2020/07/21/nfl-preseason-2020-canceled-coronavirus. Accessed May 5, 2021.

2. Mack CD, Kent RW, Coughlin MJ, et al. Incidence of Lower Extremity Injury in the National Football League: 2015 to 2018. *Am J Sports Med.* 2020;48:2287-2294.

3. Soligard T, Schwellnus M, Alonso JM, et al. How much is too much? (Part 1) International Olympic Committee consensus statement on load in sport and risk of injury. *Br J Sports Med.* 2016;50:1030-1041.

4. Bennell KL, Malcolm SA, Wark JD, Brukner PD. Models for the pathogenesis of stress fractures in athletes. *Br J Sports Med.* 1996;30:200-204.

5. Dye SF. The pathophysiology of patellofemoral pain: a tissue homeostasis perspective. *Clin Orthop Relat Res.* 2005:100-110.

6. Edwards WB. Modeling Overuse Injuries in Sport as a Mechanical Fatigue Phenomenon. *Exerc Sport Sci Rev.* 2018;46:224-231.

7. Magnusson SP, Langberg H, Kjaer M. The pathogenesis of tendinopathy: balancing the response to loading. *Nat Rev Rheumatol.* 2010;6:262-268.

8. Warden SJ, Davis IS, Fredericson M. Management and prevention of bone stress injuries in long-distance runners. *J Orthop Sports Phys Ther.* 2014;44:749-765.

9. Conte SA, Thompson MM, Marks MA, Dines JS. Abdominal muscle strains in professional baseball: 1991-2010. *Am J Sports Med.* 2012;40:650-656.

10. Erickson BJ, Gupta AK, Harris JD, et al. Rate of return to pitching and performance after Tommy John surgery in Major League Baseball pitchers. *Am J Sports Med.* 2014;42:536-543.

11. Gibson BW, Webner D, Huffman GR, Sennett BJ. Ulnar collateral ligament reconstruction in major league baseball pitchers. *Am J Sports Med.* 2007;35:575-581.

12. Makhni EC, Lee RW, Morrow ZS, Gualtieri AP, Gorroochurn P, Ahmad CS. Performance, Return to Competition, and Reinjury After Tommy John Surgery in Major

Journal Pre-proof

League Baseball Pitchers: A Review of 147 Cases. *Am J Sports Med.* 2014;42:1323-1332.

13. Marshall NE, Jildeh TR, Okoroha KR, Patel A, Moutzouros V, Makhni EC. Implications of Core and Hip Injuries on Major League Baseball Pitchers on the Disabled List. *Arthroscopy.* 2018;34:473-478.

14. Feeley BT, Kennelly S, Barnes RP, et al. Epidemiology of National Football League training camp injuries from 1998 to 2007. *Am J Sports Med.* 2008;36:1597-1603.

15. Elliott MC, Zarins B, Powell JW, Kenyon CD. Hamstring muscle strains in professional football players: a 10-year review. *Am J Sports Med.* 2011;39:843-850.

16. Orchard J, Marsden J, Lord S, Garlick D. Preseason hamstring muscle weakness associated with hamstring muscle injury in Australian footballers. *Am J Sports Med.* 1997;25:81-85.

17. Croisier JL, Ganteaume S, Binet J, Genty M, Ferret JM. Strength imbalances and prevention of hamstring injury in professional soccer players: a prospective study. *Am J Sports Med.* 2008;36:1469-1475.

18. Woods C, Hawkins RD, Maltby S, et al. The Football Association Medical Research Programme: an audit of injuries in professional football--analysis of hamstring injuries. *Br J Sports Med.* 2004;38:36-41.

19. Mair SD, Seaber AV, Glisson RR, Garrett WE, Jr. The role of fatigue in susceptibility to acute muscle strain injury. *Am J Sports Med.* 1996;24:137-143.

20. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: subjective self-reported measures trump commonly used objective measures: a systematic review. *Br J Sports Med.* 2016;50:281-291.

21. Halson SL. Monitoring training load to understand fatigue in athletes. *Sports Med.* 2014;44 Suppl 2:S139-147.

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Appendix – Data Sources Utilized

http://www.nfl.com/injuries

https://www.espn.com/nfl/injuries

https://www.cbssports.com/nfl/injuries/

https://www.pro-football-reference.com/players/injuries.htm

https://bleacherreport.com/nfl-injuries

Figure Legend

Figure 1: NFL Injures During Preseason and Regular Seasons Weeks 1-4 from 2016-2020

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Table 1: NFL Injuries During Preseason and Regular Seasons Weeks 1-4 from 2016-2020

	Year			
Week	**2016**	**2018**	**2019**	**2020**
Preseason Week 1	63	176	95	Canceled
Preseason Week 2	77	138	228	Canceled
Preseason Week 3	112	120	102	Canceled
Preseason Week 4	47	97	37	Canceled
Preseason Total	**299 (42%)**	**531 (57%)**	**462 (58%)**	**0**
Regular Season Week 1	48	100	76	141
Regular Season Week 2	177	95	93	159
Regular Season Week 3	99	94	74	136
Regular Season Week 4	89	111	95	146
Regular Season Total	**413 (58%)**	**400 (43%)**	**338 (42%)**	**582**
Total	**712**	**931**	**800**	**582**

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Table 2: NFL Injuries Per Athletic Exposure During Preseason and Regular Seasons Weeks 1-4 from 2016-2020

	Injuries Per 1000 AE		**Relative Risk Ratio**	**P Value**
	2016-2019	2020		
Preseason Week 1	2.4	Canceled	2.14	0.001
Preseason Week 2	3.2	Canceled	1.82	0.001
Preseason Week 3	2.4	Canceled	2.07	0.001
Preseason Week 4	1.9	Canceled	4.09	0.001
Regular Season Week 1	2.8	5.2	1.88	0.01
Regular Season Week 2	4.5	5.9	1.31	0.01
Regular Season Week 3	3.3	5.0	1.53	0.01
Regular Season Week 4	3.6	5.4	1.48	0.01

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Table 3: Positional Breakdown of Injuries

Positional Breakdown					
Offense	**2016**	**2018**	**2019**	**2020**	**Total**
QB	22	20	22	9	73
RB	65	97	88	45	295
WR	97	138	102	95	432
TE	62	64	66	34	226
C	14	20	14	9	57
OT	41	83	62	52	238
G	29	43	28	30	130
FB	0	2	5	3	10
Defense	**2016**	**2018**	**2019**	**2020**	**Total**
DT	90	54	95	39	278
DE	23	63	40	47	173
LB	103	119	102	79	403
CB	44	77	62	77	260
S	117	139	108	56	420

Special Teams	**2016**	**2018**	**2019**	**2020**	**Total**
K	4	12	5	6	27
LS	0	0	1	1	2

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Table 4. Injuries in NFL Football Games by Season, Anatomic Site, and Injury Mechanism

	No. of Injuries (Injuries per 1000 AEs)				
Season and Anatomic Site	**Contact**	**Noncontact**	**Overuse**	**Total**	**Injuries per 1000 AEs**
Weeks 1-4 of 2016-2017 Preseason and Regular Season					
Hip/groin	19	2	110	131	0.46
Upper leg/thigh	21	1	2	22	0.08
Knee	117	0	0	117	0.41
Lower leg/Achilles	11	9	24	44	0.15
Ankle	83	0	0	83	0.29
Foot/toes	28	0	0	28	0.10
Back	28	0	0	28	0.10
Biceps	0	0	0	0	0.00
Rib Fracture	11	0	0	11	0.04
Collarbone	1	0	0	1	0.00
Concussion	61	0	0	61	0.21
Elbow	15	0	0	15	0.05
Finger	1	0	0	1	0.00
Hand	22	0	0	22	0.08
Illness	0	6	0	6	0.02
Neck	9	0	0	9	0.03
Pectoralis Major	2	7	0	9	0.03
Toe	9	0	0	9	0.03
Shoulder	60	0	0	60	0.21

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Sports Hernia	0	0	0	0	0.00
Triceps	3	0	0	3	0.01
Undisclosed	50	0	0	50	0.18
Total	551	25	136	712	2.50

	No. of Injuries (Injuries per 1000 AEs)				
Season and Anatomic Site	**Contact**	**Noncontact**	**Overuse**	**Total**	**Injuries per 1000 AEs**
Weeks 1-4 of 2018-2019 Preseason and Regular Season					
Hip/groin	24	50	9	83	0.29
Upper leg/thigh	0	12	109	121	0.42
Knee	159	0	0	159	0.56
Lower leg/Achilles	23	6	18	47	0.16
Ankle	94	0	0	94	0.33
Foot/toes	46	0	0	46	0.16
Back	0	21	0	21	0.07
Biceps	0	2	0	2	0.01
Rib Fracture	13	0	0	13	0.05
Collarbone	0	0	0	0	0.00
Concussion	63	0	0	63	0.22
Elbow	14	0	0	14	0.05
Finger	6	5	0	11	0.04
Hand	16	0	0	16	0.06
Illness	0	16	0	16	0.06
Neck	7	0	0	7	0.02
Pectoralis Major	6	0	0	6	0.02
Toe	10	0	0	10	0.04
Shoulder	63	0	0	63	0.22

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Sports Hernia	0	0	0	0	0.00
Triceps	0	3	0	3	0.01
Forearm	4	0	0	4	0.01
Undisclosed	132	0	0	132	0.46
Total	680	115	136	931	3.26

	No. of Injuries (Injuries per 1000 AEs)				
Season and Anatomic Site	**Contact**	**Noncontact**	**Overuse**	**Total**	**Injuries per 1000 AEs**
Weeks 1-4 of 2019-2020 Preseason and Regular Season					
Hip/groin	20	22	22	64	0.22
Upper leg/thigh	13	39	40	92	0.32
Knee	108	0	0	108	0.38
Lower leg/Achilles	21	26	0	47	0.16
Ankle	81	0	0	81	0.28
Foot/toes	37	0	0	37	0.13
Back	0	23	0	23	0.08
Biceps	0	3	0	3	0.01
Rib Fracture	6	0	0	6	0.02
Collarbone	9	0	0	9	0.03
Concussion	65	0	0	65	0.23
Elbow	11	0	0	11	0.04
Finger	13	0	0	13	0.05
Hand	15	8	0	23	0.08
Illness	0	6	0	6	0.02
Neck	17	0	0	17	0.06
Pectoralis Major	6	0	0	6	0.02
Toe	6	0	0	6	0.02

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Shoulder	54	0	0	54	0.19
Sports Hernia	0	0	0	0	0.00
Triceps	0	0	0	0	0.00
Forearm	0	0	0	0	0.00
Undisclosed	129	0	0	129	0.45
Total	611	127	62	800	2.81

	No. of Injuries				
Season and Anatomic Site	**Contact**	**Noncontact**	**Overuse**	**Total**	**Injuries per 1000 AEs**
2020-2021					
Hip/groin	21	10	38	69	0.64
Upper leg/thigh	0	13	78	91	0.84
Knee	26	63	0	89	0.82
Lower leg/Achilles	10	26	0	36	0.33
Ankle	57	1	0	58	0.53
Foot/toes	20	0	0	20	0.18
Back	1	8	0	9	0.08
Biceps	3	0	0	3	0.03
Rib Fracture	18	0	0	18	0.17
Collarbone	2	0	0	2	0.02
Concussion	22	0	0	22	0.2
Elbow	9	0	0	9	0.08
Finger	15	0	0	15	0.14
Hand	5	0	0	5	0.05
Illness	0	33	0	33	0.3
Neck	8	0	0	8	0.07
Pectoralis Major	8	0	0	8	0.07
Toe	11	0	0	11	0.1
Shoulder	41	0	0	41	0.38

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Sports Hernia	0	2	0	2	0.02
Triceps	0	1	0	1	0.01
Undisclosed	32	0	0	32	0.3
Total	309	157	116	582	5.4

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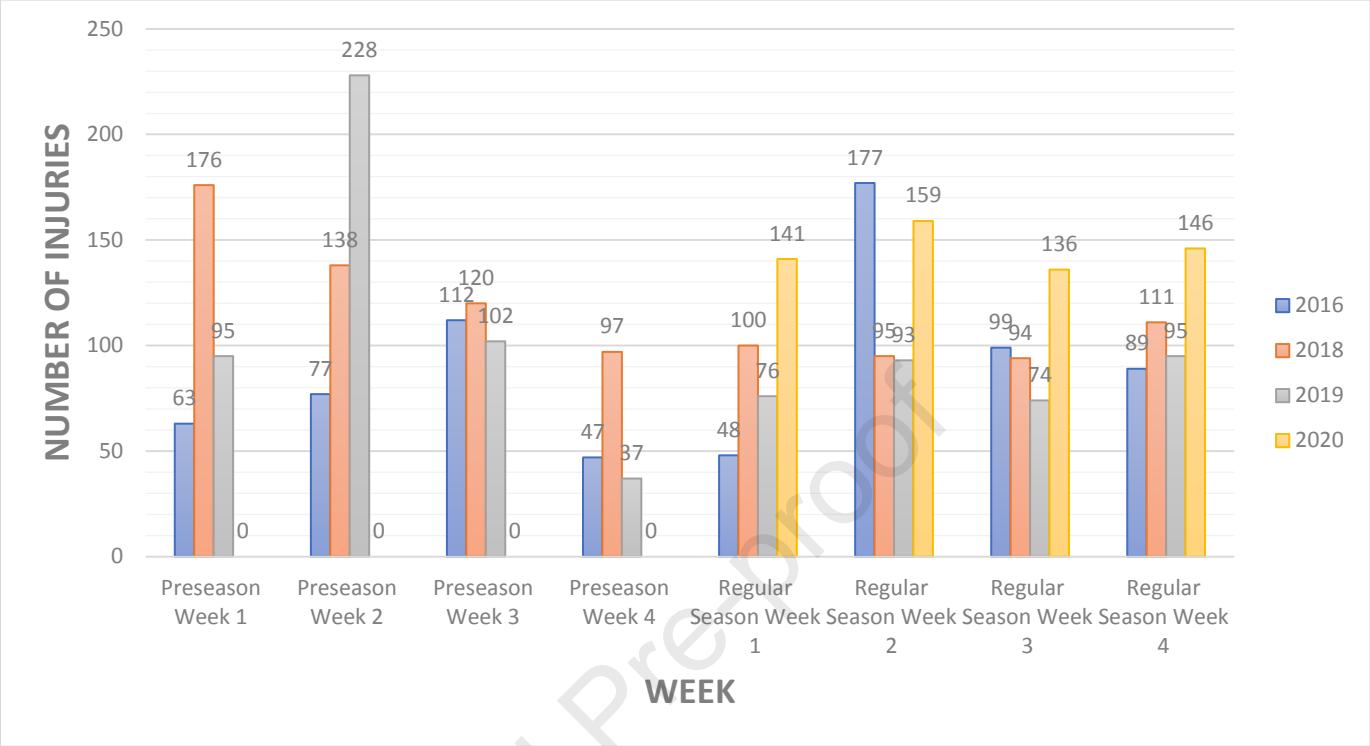
Figure 1: NFL Injuries During Preseason and Regular Seasons Weeks 1-4 from 2016-2020



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